QUINLAN A. CAMPBELL

quin@vt.edu | (919) 358-0889
LinkedIn | Personal Website | ORCID

EDUCATION

Virginia Tech, Blacksburg VA
B.S. Environmental Data Science | Minor: GIS

Relevant Coursework: Python, Intro to GIS, Modeling in GIS, Environmental Data Science, Systems Analysis, Remote Sensing, Forest Photogrammetry and Spatial Data Processing, Hydroinformatics, Biological Statistics

RESEARCH EXPERIENCE

Remote Sensing & Hydrogeodesy Student Researcher | *HIRS Lab, Virginia Tech* **Aug. 2026-Present**
-Analyze GRACE satellite data in Python to assess terrestrial water storage trends in the Chesapeake Bay region.
-Integrate water storage, drought, and groundwater datasets to evaluate groundwater change and long-term trends.

AI, Drones, and Disaster Impact Research Assistant | *Geography Department, Virginia Tech* **May 2026-Present**
-Create an ArcGIS framework for determining roads of critical importance post-disaster.
-Analyze drone and aerial imagery to assess road damage and passability following hurricanes.
-Assist in AI model validation and refinement by manually labeling image datasets.

Hydrochemistry Student Researcher | *FREC Department, Virginia Tech* **Aug. 2025-Present**
-Perform automated watershed delineation in R with LiDAR data and hydrology tools (terra, sf).
-Conduct terrain and slope analyses to classify strike- and dip-slope catchments and assess runoff.
-Support field sampling operations, including water chemistry collection and in situ sensor calibration.

Water Quality Lab Technician | *Carey Lab, Virginia Tech* **Mar. 2025-Present**
-Collect water quality samples and field observations across reservoir, inflow, and outflow sites.
-Support laboratory operations through field preparation, acid washing, data entry, and chemistry runs.
-Analyze water quality, macrophyte, and ecological monitoring data in R and ArcGIS.

Co-Lead Researcher- *Iliamna corei* | *Restoration Ecology Lab, Virginia Tech* **Aug. 2024-Present**
-Conduct fieldwork including site visits, debris clearing, and habitat assessments to support restoration.
-Develop a long-term monitoring and restoration protocol to guide future conservation actions.
-Secure grant funding to implement conservation measures.

Soybean Laboratory Technician | *Leisner Lab, Virginia Tech* **Jan. 2025-May 2025**
-Conducted comprehensive plant care, including watering and managing soybean sample processing.
-Analyzed soybean samples to investigate the impact of elevated CO_2 on nutrient levels.
-Utilized advanced tools to measure transpiration and assess leaf physiology.

BIRDNET Research Assistant | *Restoration Ecology Lab, Virginia Tech* **Jan. 2025-May 2025**
-Analyzed acoustic recordings to identify avian species presence across restored and unrestored stream sites.

Fuel Dynamics Field Research Assistant | *Catoctin Mountain & Manassas Battlefield* **May 2024-Aug. 2024**
-Assisted with field research across 340 plots in Catoctin Mountain Park and Manassas Battlefield Park.
-Conducted surveys on deadwood and trees, capturing DBH, canopy class, and decay class.

ADDITIONAL PROFESSIONAL EXPERIENCE

Disaster Divisions Intern | *United Way of Southwest Virginia* **Jan. 2026-Present**
-Maintain and standardize disaster assistance case data, including household size, case status, and funding identifiers, to support recovery operations.
-Develop geospatial visualizations in ArcGIS to analyze spatial and temporal patterns in disaster assistance, including case clustering, funding intensity, and monthly distribution across service counties.

Communications and Volunteer Engagement Intern | *Virginia Master Naturalist* **Aug. 2024-Dec. 2024**
-Collaborated with program leaders and volunteers to create a social media strategy.
-Contributed to marketing that raised awareness, recruited participants, and strengthened partnerships.
-Supported the VMN virtual conference by engaging volunteers and documenting outcomes.

Student Sustainability Intern | *Virginia Tech Office of Sustainability* **May 2024-Dec. 2024**
-Collaborated with various departments (Energy Management, Stormwater Management, Sustainable Dining, Housing & Residence Life) on technical sustainability projects as part of assigned teams.

PUBLICATIONS

Campbell, Q., Sappestein, B., Hernandez, A., Willis, B., Fitzgerald, M., Lipscomb, M., Polk, N., Reid, R. E. B., Edwards, R., Klopf, R., & Reid, J. L. (in review). Population status and trend of Iliamna corei (Peter's Mountain mallow; Malvaceae), a federally endangered Appalachian endemic.

Reid, J. L., **Campbell, Q.,** Duerr, N., Fitzgerald, H. M., Judge, P., Polk, N. D., Ripa, G. N., Sapperstein, B., Staengl, E. J., & Vrh, S. (in review). Undammed: Freeing Rivers and Bringing Communities to Life [Book review]. Restoration Ecology.

Carey, C.C., A. Breef-Pilz, D.X. Edwards, **Q. Campbell**, S.E. Kenny, and E. Walters (2026). Secchi depth data and discrete depth profiles of water temperature, dissolved oxygen, conductivity, specific conductance, photosynthetic active radiation, oxidation-reduction potential, and pH for Beaverdam Reservoir, Carvins Cove Reservoir, Falling Creek Reservoir, Gatewood Reservoir, and Spring Hollow Reservoir in southwestern Virginia, USA 2013-2025 ver 14. Environmental Data Initiative. https://doi.org/10.6073/pasta/b20ba95d3667532e4dd8bede8a00ab52

Carey, C.C., A. Breef-Pilz, D.X. Edwards, F.M. Iannucci, **Q. Campbell**, S.E. Kenny, and E. Walters (2026). Time series of dissolved methane and carbon dioxide concentrations for Falling Creek Reservoir, Beaverdam Reservoir and Carvins Cove Reservoir in southwestern Virginia, USA during 2015-2025 ver 10. Environmental Data Initiative. https://doi.org/10.6073/pasta/2b97469a6278df366478218743ab33bc

Reid, J. L., Velo, M., Aparicio-Vera, S., **Campbell, Q.**, Colberg, E., Coscia, J. T., Fitzgerald, M., Frelich, L., Hyatt, L., & Logan, C. M. (2025). Tree planting choices mediate wildfire damage to tropical forest restoration in eastern Madagascar. Restoration Ecology. https://doi.org/10.1111/rec.70251

Reid, J. L., Burt, M. A., **Campbell, Q**., Coscia, J. T., Huston, J. C., Kubesch, J. O. C., MacDougal, P., Ripa, G. N., Staengl, E. J., & Viani, R. A. G. (2024). Ecological Restoration: Moving Forward Using Lessons Learned [Book review]. Restoration Ecology. https://doi.org/10.1111/rec.14163

PRESENTATIONS

Campbell, Q., Sapperstein, B., Reid, J.L., *Population Status and Trend of Iliamna corei (Peter's Mountain Mallow; Malvaceae), a Federally Endangered Appalachian Endemic.* Poster presentation, School of Plant and Environmental Science Symposium, Blacksburg, VA, 2026.

Campbell, Q., Sapperstein, B., Reid, J.L., *Population Status and Trend of Iliamna corei (Peter's Mountain Mallow; Malvaceae), a Federally Endangered Appalachian Endemic.* Poster presentation, Dennis Dean Undergraduate Research and Creative Scholarship Conference, Blacksburg, VA, 2026.

Campbell, Q., Gannon, J.P., *Hydrochemical Signatures of Scarp and Dip-Slope Drainage in the Valley and Ridge Region.* Poster presentation, Geological Society of America Joint Southeastern, North-Central, and South-Central Section Meeting, Memphis, TN, 2026.

Campbell, Q., Sapperstein, B., Reid, J.L., *Conservation and Management Procedures for Critically Endangered Peter's Mountain Mallow (Iliamna corei)*. Poster presentation, Biological and Ecological Restoration Initiative Symposium, Blacksburg, VA, 2025.

GRANTS AND ACADEMIC HONORS

Virginia Tech Drone Park Part 107 Grant: $175 (2026)
College of Natural Resources and Environment Undergraduate Research Fellowship: $2,500 (2026)
Ut Prosim Internship Support Fund: $800 (2026)
Office of Undergraduate Research Travel Grant: $487.98 (2026)
Global Change Center Undergraduate Research Grant: $316 (2024)
Dean's List (Spring & Fall 2024, Spring & Fall 2025, Spring 2026)

LEADERSHIP, ACTIVITIES, SERVICE, VOLUNTEER WORK

Service Chair, Philanthropy Chair, Pledge Parent | Lambda Iota Mu (Co-ed Service Frat) (Feb. 2024-Present)
Coordinating and leading service, philanthropy, and new-member initiatives.
Volunteer | Campus Kitchen (Nov. 2025-Present)
Coordinating the collection of surplus food from campus dining facilities and transporting it to local food banks.

PROFESSIONAL MEMBERSHIPS AND AFFILIATIONS

Ecological Society of America (ESA)
American Geophysical Union (AGU)
Geological Society of America (GSA)

CERTIFICATIONS

FAA Part 107 Remote Pilot Certificate
FEMA IS-100.c - Introduction to the Incident Command System
FEMA IS-700.b - National Incident Management System (NIMS)
Yoga Alliance 200-hour Yoga Teacher Training